Adaptation Experiment 4: TS-IFA

1 Theoretical overview and goal

TS-IFA is a trainable adapter around a frozen forecast backbone. It forms four candidates: vanilla prediction $\hat{y}_0$, context-conditioned prediction $\hat{y}_c$, a residual-attention prediction $\hat{y}_r$, and a memory-attention prediction $\hat{y}_m$. Cross-attention reads retrieved examples and the final forecast is a horizon-wise convex mixture

$$\hat{y}_h = \sum_{b\in\{0,c,r,m\}} \pi_{b,h}\hat{y}_{b,h}, \qquad \pi_{b,h} = \mathrm{softmax}_b(a_{b,h}).$$

Residual attention uses neighbor backbone errors as values; memory attention uses neighbor futures. A third attention block produces mixture logits from query and neighbor look-backs. The horizon-specific four-way mixture is the highest-capacity final component and is the first target for simplification if T_2 overfitting persists.

Experimental goal. Learn a small, backbone-agnostic adaptation mechanism that can exploit retrieved forecasts, outcomes, and errors while remaining anchored to the reliable vanilla forecast.

2 Objective and initialization

With nMSE scaling, training minimizes

$$\mathcal{L} = \mathcal{L}_{\mathrm{pred}} + \beta\|\hat{y} - \hat{y}_0\|_{\mathrm{n}}^2 + \gamma\big(\|\Delta_r - (y - \hat{y}_0)\|_{\mathrm{n}}^2 + \|\Delta_m - (y - \hat{y}_0)\|_{\mathrm{n}}^2\big).$$

Correction heads are zero-initialized. Non-vanilla mixture logits start at -6, so the initial adapter is close to the backbone and learns deviations only when supported by data.

3 Implementation details

Extraction stores pooled $T_1 + T_2$ rather than prescribing their boundary. TS-IFA assigns the final 20% of whole query dates in that pool to T_2. This model-local split uses the same $X_s = (s - L, s]$, $Y_s = (s, s + H]$ convention as every baseline.

Each attention block uses four heads of dimension 32, followed by a 128-unit GELU MLP. Inputs are instance-normalized on the query scale. AdamW uses learning rate 10^{-5}, weight decay 10^{-4}, gradient clipping 1, $\beta = \gamma = 10^{-2}$, batch size 256, and 10,000 epochs. Every epoch draws one random batch from T_1. Full deterministic T_2 validation and interval logging occur every 1,000 optimizer steps; T_3 is evaluated only after training.

The trainer restores the lowest-nMSE T_2 checkpoint before touching T_3 and can stop after a configurable number of non-improving validation evaluations. The launcher enables best-checkpoint restoration but leaves early stopping disabled by default for backward-compatible optimization length. Unlike ridge and fixed-candidate CatBoost gates, TS-IFA is not refit on $T_1 + T_2$ after checkpoint selection: doing so would remove the validation signal used to select its weights. Dropout and reduced attention/MLP dimensions should be screened on one pilot configuration before widening the sweep. A scalar/shared or low-rank final mixture is the next architecture ablation, not yet a default change.

The small sweep covers Traffic/Electricity/Solar and the four RevIN reproduction settings with raw/instance Euclidean retrieval, $k \in \{1, 3, 10\}$, and Chronos. Full adds ETTh1 (all seven variables), Weather, Exchange, and 512–64 while retaining Chronos; ultra adds TabPFN-TS after the Chronos architecture is stable. Implementation is under `src/adaptors/ts_ifa/`; the launcher is `ts_ifa.slurm`. Saved artifacts include the model, resolved configuration, full step history, interval-average train and T_2 validation plot, T_3 metrics, and branch predictions.

4 Interpretation

Report TS-IFA beside vanilla, context, residual, and memory branches. This distinguishes gains from better candidates from gains due to mixture learning. Selection must use T_2; the untouched T_3 period is the only final test. Attention weights are diagnostics and should not be interpreted as causal feature importance.